

RDD: Robust Feature Detector and Descriptor using Deformable Transformer

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논문 링크: <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=11093965>

Git-Hub 코드 링크: <https://xtcpete.github.io/rdd/>.

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1. Abstract & Introduction

Viewpoint (시점), **Scale** (이미지안에서의 대상 크기), **Illumination** (같은 대상이 있는 이미지의 빛의 조건) 변화와 같은 극적인 변동 속에서도 신뢰할 수 있는 **Keypoint**와 **Descriptor**가 필요

문제 제기

Rule-based
(SIFT · SURF · ORB)

Handcrafted 방식으로 까다로운 조건에서 성능 저하.
쉽게 말해 사람이 다 정의해야 해서 변동성 흡수 어려움

Learning-based
(SuperPoint · DISK)

Local Kernel로 Long-range 관계 포착 불가,
Geometric Invariance 미반영

Deformable Conv
(ASLFeat · ALIKED)

임의의 기하 변형을 모델링 하나, **Local Window** 안으로 제한되어 **Vanishing Line**처럼 Long Range에 의존하는 **Visual** 학습 어려움

제안하는 방법 (RDD)

Deformable Attention Descriptor

Query마다 소수의 핵심 위치만 **Sampling** Long-range · Geometric Invariance 학습

Two-branch 구조

Keypoint는 CNN, **Descriptor**는 Transformer 두 목표가 달라 분리 학습 → 수렴·성능 향상

Semi-dense Matching (RDD*)

Sparse Match(RDD)의 **Epipolar Constraint**로 거친 **Semi-dense Match**의 위치를 보정 (RDD*)

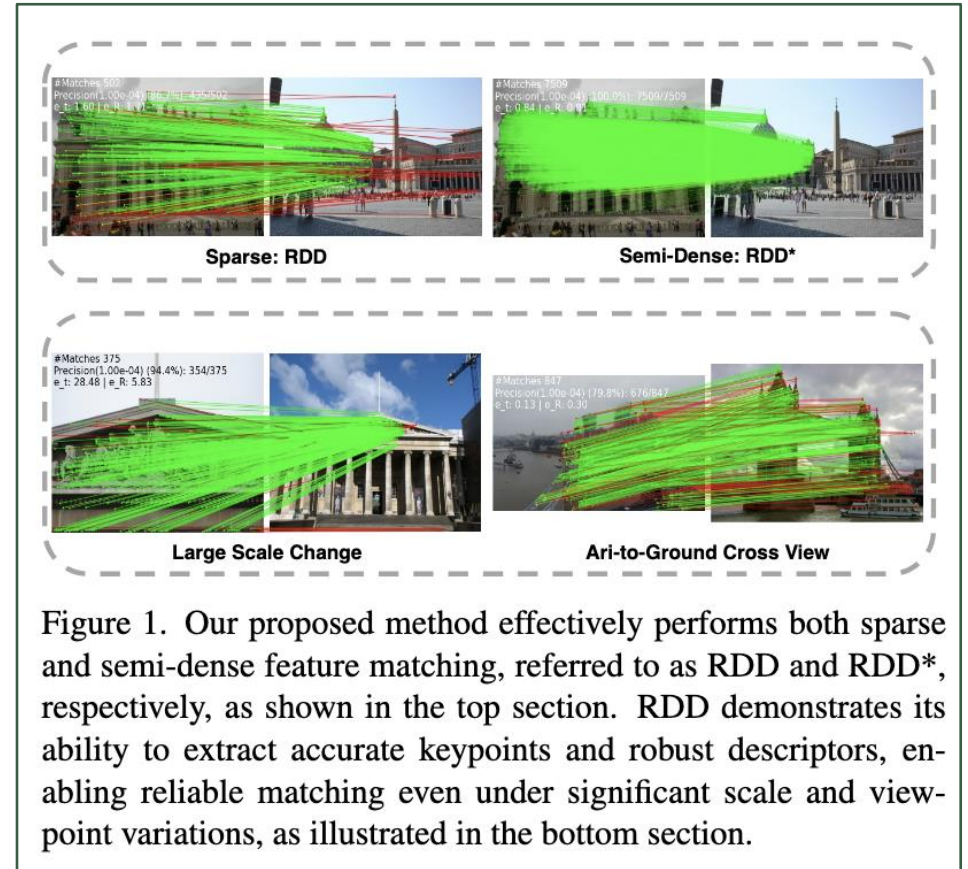
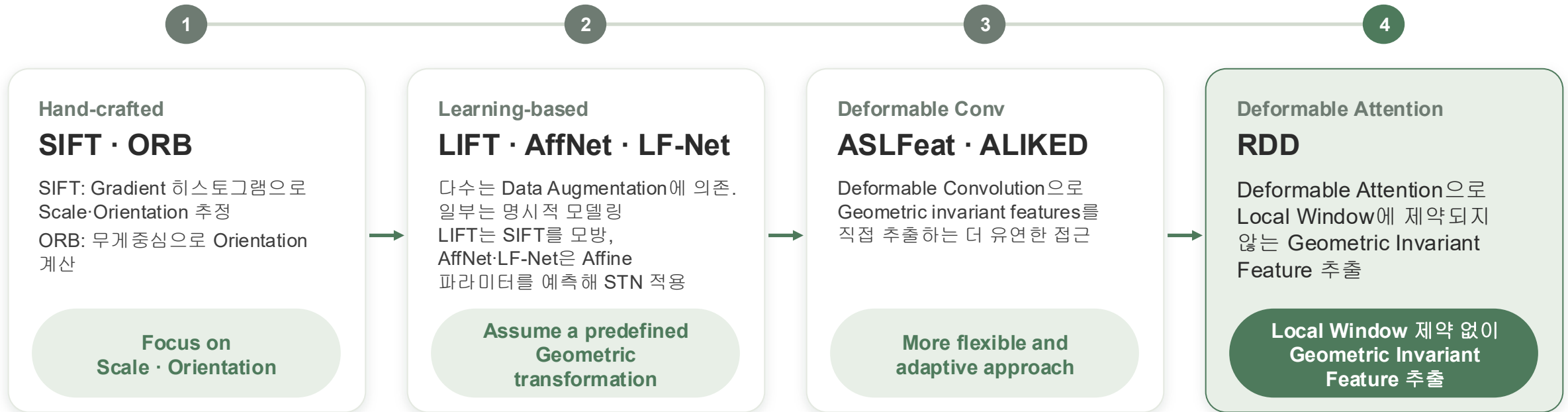


Figure 1. Our proposed method effectively performs both sparse and semi-dense feature matching, referred to as RDD and RDD*, respectively, as shown in the top section. RDD demonstrates its ability to extract accurate keypoints and robust descriptors, enabling reliable matching even under significant scale and view-point variations, as illustrated in the bottom section.

RDD: Robust Deformable Detector (**RDD***: Semi-dense Matching 버전), **SIFT**: Scale-Invariant Feature Transform, **SURF**: Speeded-Up Robust Features, **ORB**: Oriented FAST and Rotated BRIEF, **CNN**: Convolutional Neural Network, **SuperPoint**: Self-Supervised Interest Point Detection and Description, **DISK**: Learning Local Features with Policy Gradient, **ASLFeat**: Learning Local Features of Accurate Shape and Localization, **ALIKED**: A Lighter Keypoint and Descriptor Extraction Network via Deformable Transformation

● 2. Related Work – 2.1. Geometric Invariant Descriptors

Geometric Invariant Descriptor는 Hand-crafted에서 Learning-based를 거쳐 Deformable Attention으로 발전



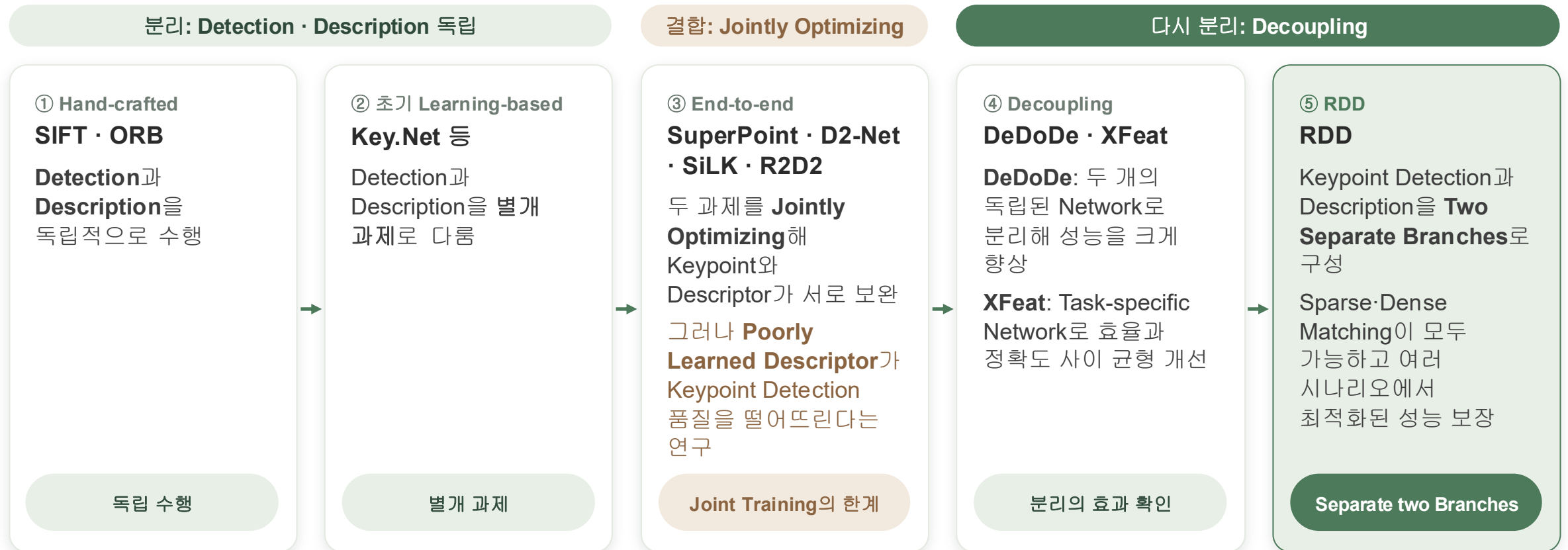
RDD는 Deformable Convolution에서 영감을 받아 Deformable Attention을 쓴다

Deformable Attention으로 **Local Window**에 제한되지 않는 **Geometric Invariant Feature**를 추출한다.

SIFT: Scale-Invariant Feature Transform, **ORB**: Oriented FAST and Rotated BRIEF, **LIFT**: Learned Invariant Feature Transform, **AffNet**: Learning Affine Regions via Discriminability, **LF-Net**: Learning Local Features from Images, **STN**: Spatial Transformer Networks, **ASLFeat**: Learning Local Features of Accurate Shape and Localization, **ALIKED**: A Lighter Keypoint and Descriptor Extraction Network via Deformable Transformation, **RDD**: Robust Deformable Detector

● 2. Related Work – 2.2. Decoupling Keypoint and Descriptor Learning

Detection과 **Description**을 분리 수행하던 과거의 방식 대비 **Joint Training**이 서로를 보완한다고 생각했으나, 오히려 한쪽이 **Poor**하면 둘다 품질이 떨어진다는 것을 확인하고 다시 **Decoupling** 해서 수행



SIFT: Scale-Invariant Feature Transform, **ORB**: Oriented FAST and Rotated BRIEF, **Key.Net**: Keypoint Detection by Hand-crafted and Learned CNN Filters, **SuperPoint**: Self-Supervised Interest Point Detection and Description, **D2-Net**: A Trainable CNN for Joint Detection and Description of Local Features, **SiLK**: Simple Learned Keypoints, **R2D2**: Repeatable and Reliable Detector and Descriptor, **DeDoDe**: Detect, Don't Describe—Describe, Don't Detect for Local Feature Matching, **XFeat**: Accelerated Features for Lightweight Image Matching, **RDD**: Robust Deformable Detector

● 2. Related Work – 2.3. Efficient Attention Mechanisms

Deformable Attention은 고정된 개수의 **Pixel**만 참조해 **Global Context**를 효율적으로 학습

문제 제기 **Multi-head Attention**

계산 복잡도가 **Quadratic** → 학습 일정(Training Schedule)이 길어지고, **Spatial Resolution**을 제한해야 한다

효율화 접근

Local Attention

LR-Net

- 2D 이미지용 **Local Relation Layer** 제안
- Attention을 고정된 **Local Window** 안에서만 계산

Window: 고정 · Local

Kernel 대체

Linear Transformer

- Exponential Kernel을 다른 **Kernel Function**으로 대체
- Vanilla Attention의 계산 복잡도를 **Linear**로 감소

복잡도: Linear

Query 의존 Window

Deformable Attention

- Attention Window가 **Query Feature**에 따라 달라짐
- Linear Projection이 **Displacement** 집합과 그 가중치 예측
- 각 Pixel이 임의 거리의 고정된 개수 Pixel에 Attention

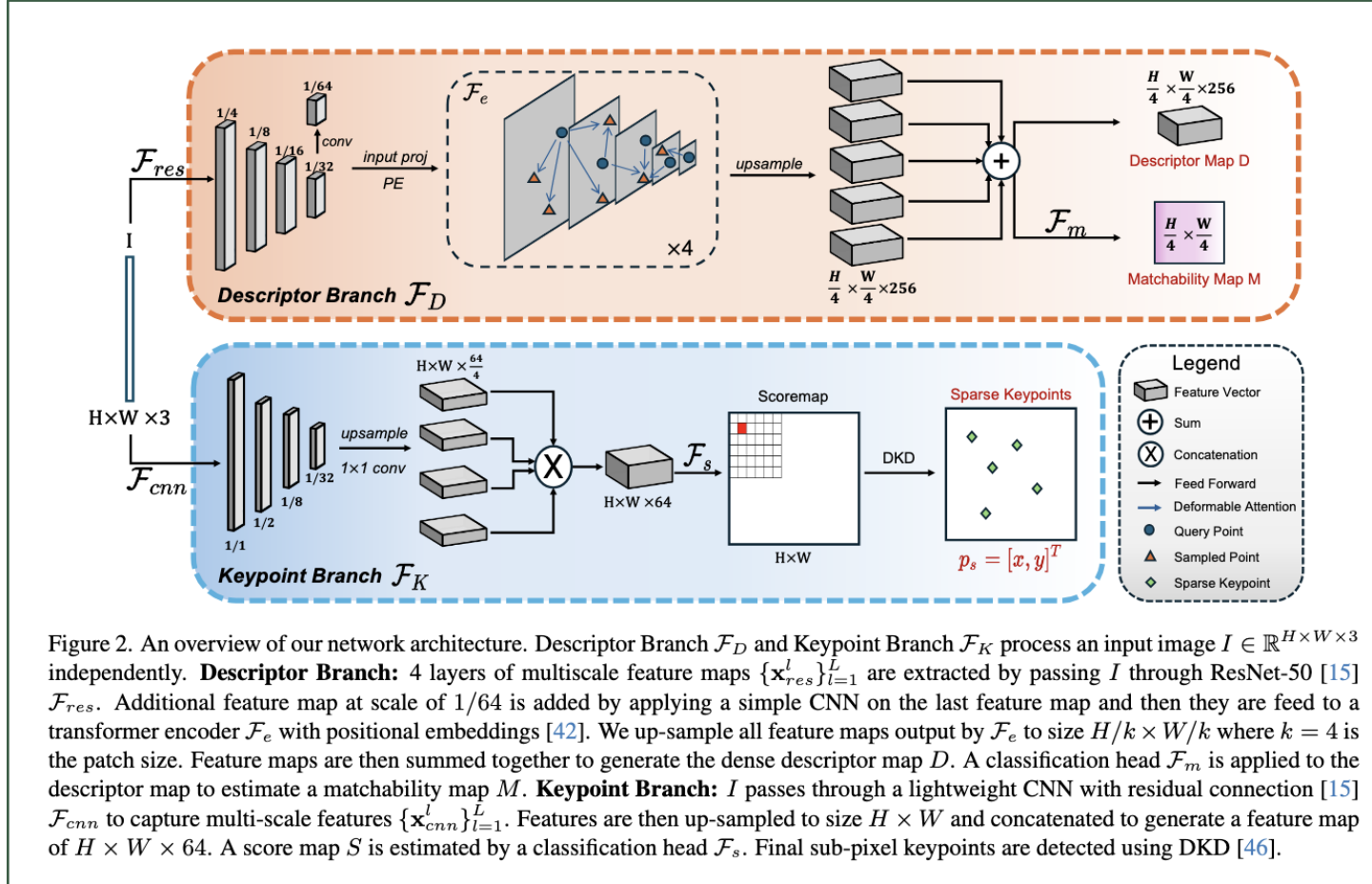
Window: Query 의존 · 임의 거리

Deformable Attention은 **Global Context**를 효율적으로 학습할 수 있어서 논문의 저자들이 적합한 방법으로 채택

LR-Net: Local Relation Networks, **RDD**: Robust Deformable Detector

● 3. Methods – Overview

RDD는 Descriptor Branch와 Keypoint Branch가 독립적으로 수행(Process)



Descriptor Branch $\mathcal{F}_D$

- 입력 $I \in \mathbb{R}^{H \times W \times 3}$ 에서 ResNet-50($\mathcal{F}_{res}$)으로 Multi-scale Feature Map $\{x_{res}^l\}_{l=1}^L$ 추출
- Deformable Transformer Encoder $\mathcal{F}_e$ 로 Descriptor Map $D \in \mathbb{R}^{H/k \times W/k}$ 추정 ($k=4$: Patch Size)
- Classification Head $\mathcal{F}_m$ 으로 D 에서 Matchability Map $M \in \mathbb{R}^{H/k \times W/k}$ 추정

Keypoint Branch $\mathcal{F}_K$

- Residual Connection을 가진 경량 CNN $\mathcal{F}_{cnn}$ 과 Classification Head $\mathcal{F}_s$ 로 Scoremap $S \in \mathbb{R}^{H \times W}$ 생성
- DKD로 Sub-pixel Keypoint $p_s = [x, y]^T$ 검출
- 검출된 Keypoint로 D 에서 Descriptor $d \in \mathbb{R}^{256}$ Sampling

ResNet: Deep Residual Learning for Image Recognition, **CNN:** Convolutional Neural Network, **PE:** Positional Embedding, **DKD:** Differentiable Keypoint Detection, **RDD:** Robust Deformable Detector

● 3. Methods – 3.1. Preliminaries

RDD는 Sub-pixel 검출에 DKD를, Descriptor 추출에 Multi-scale Deformable Attention을 사용

Differentiable Keypoint Detection (DKD)

Keypoint Branch

① NMS로 Integer Coordinate 추출 ($N \times N$ Window, $N=5$)

$$p_{nms} = [x', y']^T$$

NMS

② Window 내 Score를 Softmax로 확률화

$$s'(i,j) = \text{softmax}((s(i,j) - s_{\max}) / t_{\det}) \quad (1)$$

③ Soft-argmax로 Expected Location 계산

$$[\hat{i}, \hat{j}]_{\text{soft}} = \sum s'(i,j) \cdot [i, j]^T \quad (2)$$

④ Sub-pixel Keypoint = Integer Coordinate + Soft Offset

$$p_s = [x', y']^T + [\hat{i}, \hat{j}]_{\text{soft}} \quad (3)$$

Multi-scale Deformable Attention

Descriptor Branch

$$x_{\text{msd}}(z_q, \hat{p}_q, \{x^l\}) = \sum_m W_m A'_{mq} \quad (6)$$

$$A'_{mq} = \sum_l \sum_k A_{mlqk} \cdot W'_m x_l(f_s(\hat{p}_q) + \Delta p_{mlqk}) \quad (7)$$

기호	의미
m	Attention Head Index
k	Sampled Key Index (총 K 개)
l	Feature Map Scale Level (총 L 개)
Δp_{mlqk}	k 번째 Sampling Point의 Sampling Offset
A_{mlqk}	k 번째 Sampling Point의 Attention Weight
f_s	정규화 좌표 p 를 l 번째 Level Feature Map 크기로 Re-scale

Δp 와 A 는 Query Feature에서 Linear Layer로 예측

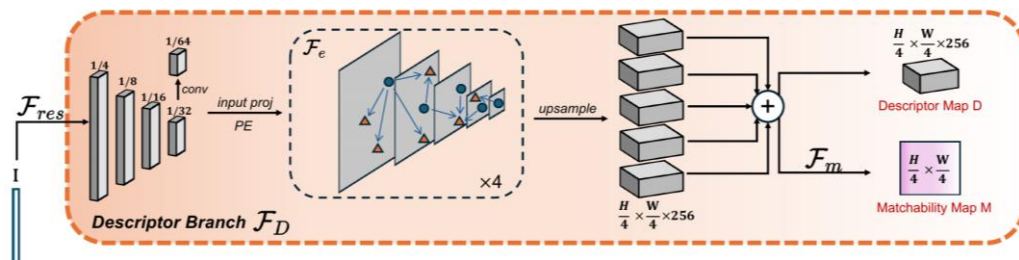
DKD: Differentiable Keypoint Detection, NMS: Non-Maximum Suppression, RDD: Robust Deformable Detector

● 3. Methods – 3.2. Network Architecture

Descriptor는 넓게 보는 transformer로, Keypoint는 정밀한 CNN으로 따로 수행

BRANCH 1 Descriptor Branch F_D

Dense Feature Map D 와 Matchability Map M 추출



① ResNet-50 F_{res} : $1/4 \cdot 1/8 \cdot 1/16 \cdot 1/32$ Scale 추출

② 마지막 Map에 CNN 적용 → **1/64 Scale** 추가

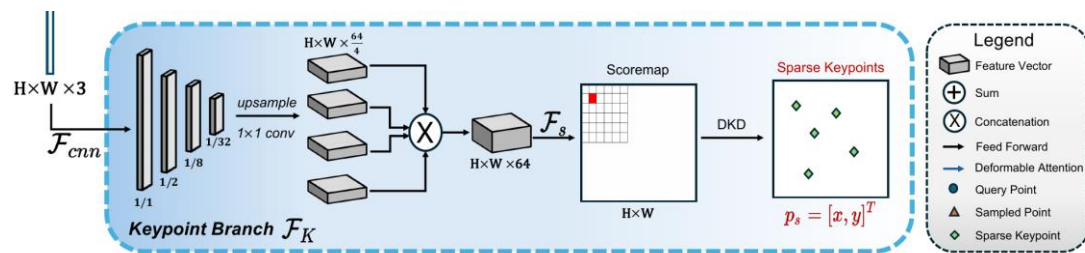
③ **Positional Encoding** 후 F_e 에 입력

④ **Multi-scale Deformable Attention**: 4 Layer $\times$ 8 Head $\times$ 8 Point

⑤ 출력: $D \in \mathbb{R}^{H/4 \times W/4 \times 256}$, Matchability Map M

BRANCH 2 Keypoint Branch F_K

Descriptor 추정 영향 없이 정확한 Keypoint 검출에만 전념



① 경량 CNN F_{cnn} : $1/1 \cdot 1/2 \cdot 1/8 \cdot 1/32$ Scale (32-dim)

② 원본 해상도로 **Upsample** → **Concat**: $H \times W \times 128$

③ F_s 로 **Score Map** 추정

④ **DKD**: NMS 후 Sub-pixel Keypoint 검출

ResNet: Deep Residual Learning for Image Recognition, CNN: Convolutional Neural Network, DKD: Differentiable Keypoint Detection, NMS: Non-Maximum Suppression

● 3. Methods – 3.3. Feature Matching(1/2)

RDD*는 Sparse Match로 구한 Epipolar Constraint로 Coarse Match 보정

공통 입력 Single Forward Pass로 p_s, D, M 출력 $\rightarrow D$ 를 원본 해상도로 Bilinear Upsample $\rightarrow p_s$ 위치에서 Descriptor d Sampling

Sparse Matching

RDD · Dual-Softmax

$$S(i,j) = (1/\tau) \cdot \langle d_i^1, d_j^2 \rangle$$

$$P(i,j) = \frac{\text{softmax}(S(i,\cdot))_j}{\text{softmax}(S(\cdot,j))_i} \quad (8)$$

- τ : Temperature
- S 의 두 차원 모두에 Softmax $\rightarrow$ Soft MNN Matching 확률 P

Confidence > 0.01인 쌍만
Sparse Match m_s 로 채택

Semi-dense Matching

RDD* · Sparse Match가 Guide하는 Refinement (제안 Module)

① Coarse Match

M 기준 Top-K Coarse Keypoint $p_c \rightarrow$ 식 (8)로 Coarse Match m_c (1/4 해상도라 보정 필요)



② Fundamental Matrix 추정

Sparse Match m_s 로 Eight-point Method $\rightarrow F$ 추정



③ Epipolar Line (9)

p_c^1 은 고정, Homogeneous 좌표 p_h^1 로 $L = F \cdot p_h^1$ 계산 (계수 a, b, c)



④ Offset 계산 · Outlier 제거 (10)–(11)

Epipolar Constraint를 만족하는 $(\Delta x, \Delta y)$ 로 p_c^2 보정 · Offset > Patch Size(4)면 제거

MNN: Mutual Nearest Neighbor, RDD: Robust Deformable Detector (Sparse), RDD*: Semi-dense Matching 버전

● 3. Methods – 3.3. Feature Matching(2/2)

RDD*는 Epipolar Line 위의 닫힌 해로 Coarse Match를 Sub-pixel로 보정

Epipolar Refinement 수식

③ Epipolar Line

(9)

$$L = F \cdot p_h^1$$

p_c^1 을 Homogeneous 좌표 p_h^1 로 변환 · 각 Line은 계수 a, b, c

④-1 Epipolar Constraint

(10)

$$a \cdot (x + \Delta x) + b \cdot (y + \Delta y) + c = 0$$

이 선형 방정식을 풀어 Constraint를 강제

④-2 Offset (Closed-form)

(11)

$$\Delta x = [b(bx - ay) - ac] / (a^2 + b^2 + \varepsilon) - x$$

$$\Delta y = [a(ay - bx) - bc] / (a^2 + b^2 + \varepsilon) - y$$

(x, y) : p_c^2 의 Pixel 좌표

Offset > Patch Size(4) → Matched Patch 밖으로 이동한 Outlier로 제거

최종 Refined Dense Match $m_c = (x + \Delta x, y + \Delta y)$

기존 Refinement와의 차이

기존 방법 (ASpanFormer, LoFTR)

Fine Feature Crop → Offset 예측

- Coarse Match 주변 Fine-level Feature를 I^1, I^2 에서 Crop
- I^2 에서 Offset을 예측

RDD* (Ours)

Simple · Efficient · Accurate Module

- 선행 연구(Geometry-aware Feature Matching)에서 영감: Sparse Correspondence가 Semi-dense Matching을 Guide
- 해당 연구는 Detector-free 방법의 Coarse Match를 반복 재할당, RDD*는 같은 아이디어로 Coarse → Fine Refine
- p_c^1 은 고정, p_c^2 만 Epipolar Constraint로 보정

Refinement를 빼면 (Table 5, Sec. 4.4)

RDD* MegaDepth AUC@5° 51.3 → 41.3

RDD*: Semi-dense Matching 버전, AUC: Area Under Curve

● 3. Methods – 3.4. Implementation Details

Descriptor Branch를 먼저 학습한 뒤 Freeze하고 Keypoint Branch를 학습

TRAINING DATA

Pseudo Ground Truth 생성

- **MegaDepth + Air-to-Ground**
- Air-to-Ground: 인터넷 Drone 영상과 유명 Landmark의 Ground 이미지로 수집
- **Pose와 Depth**로 Pixel 대응 생성

$$m_{gt} \in \mathbb{R}^{N \times 4}$$

- 앞 2열: I 의 Pixel 좌표
- 뒤 2열: Warping 함수 f_{warp} 로 얻은 I' 의 좌표

두 Branch는 따로 학습

Descriptor → Freeze → Keypoint

Training Descriptor Branch

LOSS L_D

STEP 1 · Focal Loss (12)

$$L_{focal} = -\alpha(1 - P_{ij})^\gamma \cdot \log(P_{ij})$$

Descriptor 감독 · 식 (8)의 대각 P_{ii} (Positive)를 1로 · $\alpha = 0.25$, $\gamma = 2$

STEP 2 · BCE Loss (13)

$$L_{BCE} = -(M_{gt} \log M + (1 - M_{gt}) \log(1 - M))$$

m_{gt}/k 로 GT Matchability Map M_{gt} 생성

STEP 3 · Matchability Loss (14)

$$\lambda = e^{-L_{BCE}}$$

$$L_{matchability} = -\alpha(1 - \lambda)^\gamma \cdot L_{BCE}$$

식 (12)의 P_{ii} 를 λ 로 대체한 Modified Focal Loss

STEP 4 · 최종 Loss · 학습 설정

$$L_D = L_{focal} + L_{matchability}$$

800 Resize · Batch 32 · AdamW · LR 1e-4 (20k Step 후 2k마다 $\times 0.5$) · H100 8장, 1일

Training Keypoint Branch

LOSS L_K · Descriptor Branch Freeze 후 학습

STEP 1 · Reprojection Loss (15)

$$L_{reprojection} = \frac{1}{2}(\text{dist}^{1 \rightarrow 2} + \text{dist}^{2 \rightarrow 1})$$

f_{warp} 로 옮긴 Keypoint와 5 Pixel 미만 최근접점의 거리

STEP 2 · Reliability Loss (16)–(18)

$$R = \exp((P - 1) / t_{rel})$$

$$L_{reliability} = \frac{1}{2}(L^1_{reliability} + L^2_{reliability})$$

Score $s^1 \cdot s^{1 \rightarrow 2}$ 가중 · $(1 - r^1)$ 최소화

STEP 3 · Dispersity Peakiness Loss (19)–(20)

$$L_{peaky} = (1/N^2) \sum d(i,j) \cdot s'(i,j)$$

$$d(i,j) = \|[i,j] - [\hat{i},\hat{j}]_{soft}\|_p$$

식 (3)의 Soft 항과 NMS 항의 최적화 방향 정렬

STEP 4 · 최종 Loss · 학습 설정

$$L_K = L_{reprojection} + L_{reliability} + L_{peaky}$$

DKD $N = 5$, Top 500 + Non-salient 500 Random · H100 1장, 4시간

GT: Ground Truth, BCE: Binary Cross-Entropy, NMS: Non-Maximum Suppression, DKD: Differentiable Keypoint Detection

● 4. Experiments – 4.1. Relative Pose Estimation (1/2)

매칭이 어려울수록 2위 대비 **AUC@5°** 격차 확대: MegaDepth-1500 +1.0 → MegaDepth-View +5.2 → Air-to-Ground +9.9pt

DATASET

- **MegaDepth-1500**: D2-Net의 표준 Outdoor Pose 벤치마크, 같은 Test Split
- **MegaDepth-View** (신규): MegaDepth Test Scene 중 Viewpoint-Scale 변화가 큰 Pair
- **Air-to-Ground** (신규): 수집 데이터에서 1,500 Pair Sampling. Large-baseline Cross-view

METRICS & COMPARING

- 복원 Pose의 **AUC** @5°·10°·20°
- **RANSAC**으로 Essential Matrix 추정
- MegaDepth: Detector/Descriptor 및 Learned Matching 방법과 비교
- Air-to-Ground: Detector/Descriptor 방법과 비교
- Sparse 방법: 긴 변 1,600px, Top 4,096 Feature

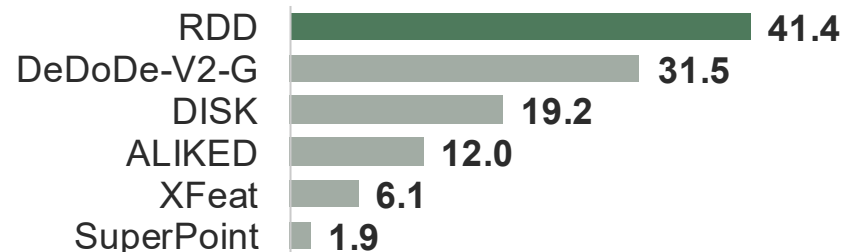
Table 1. **SotA comparison on the MegaDepth [22]**. Results are measured in AUC (higher is better). Top 4,096 features used to all sparse matching methods. Best in bold, second best underlined.

Method	MegaDepth-1500			MegaDepth-View		
	@5°	@10°	@20°	@5°	@10°	@20°
Dense						
DKM [9] CVPR'23	60.4	74.9	85.1	67.4	80.0	88.2
RoMa [12] CVPR'24	62.6	76.7	86.3	69.9	81.8	89.4
Semi-Dense						
LoFTR [39] CVPR'21	52.8	69.2	81.2	50.6	65.8	77.9
ASpanFormer [5] ECCV'22	55.3	71.5	83.1	61.3	75.2	84.6
ELoFTR [44] CVPR'24	56.4	72.2	83.5	60.2	74.8	84.7
XFeat* [32] CVPR'24	38.6	56.1	70.5	23.9	37.7	51.8
RDD*	51.3	67.1	79.3	41.5	55.7	66.7
Sparse with Learned Matcher						
SP [6]+SG [37] CVPR'19	49.7	<u>67.1</u>	<u>80.6</u>	51.5	65.5	76.9
SP [6]+LG [24] ICCV'23	<u>49.9</u>	67.0	80.1	<u>52.4</u>	<u>67.3</u>	<u>78.5</u>
Dedode-V2-G [10, 11]+LG [24] ICCV'23	44.1	62.1	76.5	41.9	57.1	70.0
RDD+LG [24] ICCV'23	52.3	68.9	81.8	54.2	69.3	80.3
Sparse with MNN						
SuperPoint [6] CVPRW'18	24.1	40.0	54.7	<u>7.50</u>	13.3	21.1
DISK [41] NeurIPS'20	38.5	53.7	66.6	30.4	41.9	51.6
ALIKED [47] TIM'23	41.8	56.8	69.6	30.0	41.8	53.2
XFeat [32] CVPR'24	24.0	40.1	55.8	8.82	16.5	26.8
DeDoDe-V2-G [10, 11] CVPRW'24, 3DV'24	<u>47.2</u>	<u>63.9</u>	<u>77.5</u>	<u>33.1</u>	<u>47.6</u>	<u>60.2</u>
RDD	48.2	65.2	78.3	38.3	53.1	65.6

Table 2. **SotA comparison on proposed Air-to-Ground benchmark**. Keypoints and descriptors are matched using dual-softmax MNN. Measured in AUC (higher is better). Best in bold, second best underlined.

Method	@5°	@10°	@20°
SuperPoint [6] CVPRW'18	1.89	3.56	6.81
DISK [41] NeurIPS'20	19.2	27.1	34.6
ALIKED [47] TIM'23	12.0	17.8	25.8
XFeat [32] CVPR'24	6.12	11.4	17.5
DeDoDe-V2-G [10, 11] CVPRW'24, 3DV'24	<u>31.5</u>	<u>45.3</u>	<u>58.3</u>
RDD	41.4	56.0	67.8

Table 2 · Air-to-Ground, AUC@5° (Dual-softmax MNN)



수치의 의미: 복원한 Relative Pose의 AUC(%)를 Threshold 5°·10°·20°에서 측정, 높을수록 좋음 · Bold = 1위, Underline = 2위 · Table 1은 모든 Sparse 방법에 Top 4,096 Feature 사용, Table 2는 Dual-softmax MNN 매칭
SP: SuperPoint, **SG**: SuperGlue, **LG**: LightGlue (Learned Matcher), **RANSAC**: Random Sample Consensus, *: Semi-dense 버전 (RDD*, XFeat*)

● 4. Experiments – 4.1. Relative Pose Estimation (2/2)

큰 Scale·Viewpoint 변화에서 RDD* · RDD가 DeDoDe-V2-G·ALIKED보다 Robust한 Matching 수행

RESULTS (Table 1, 2 · Fig. 3)

- **Sparse 설정:** SOTA DeDoDe-V2-G 보다 우수하며, 어려운 Scenario일수록 격차가 큼
- **RDD + LightGlue**도 기존 방법보다 우수
- **MegaDepth-View · Air-to-Ground:** 큰 Viewpoint·Scale 변화에서 기존 모든 Detector/Descriptor 방법보다 우수

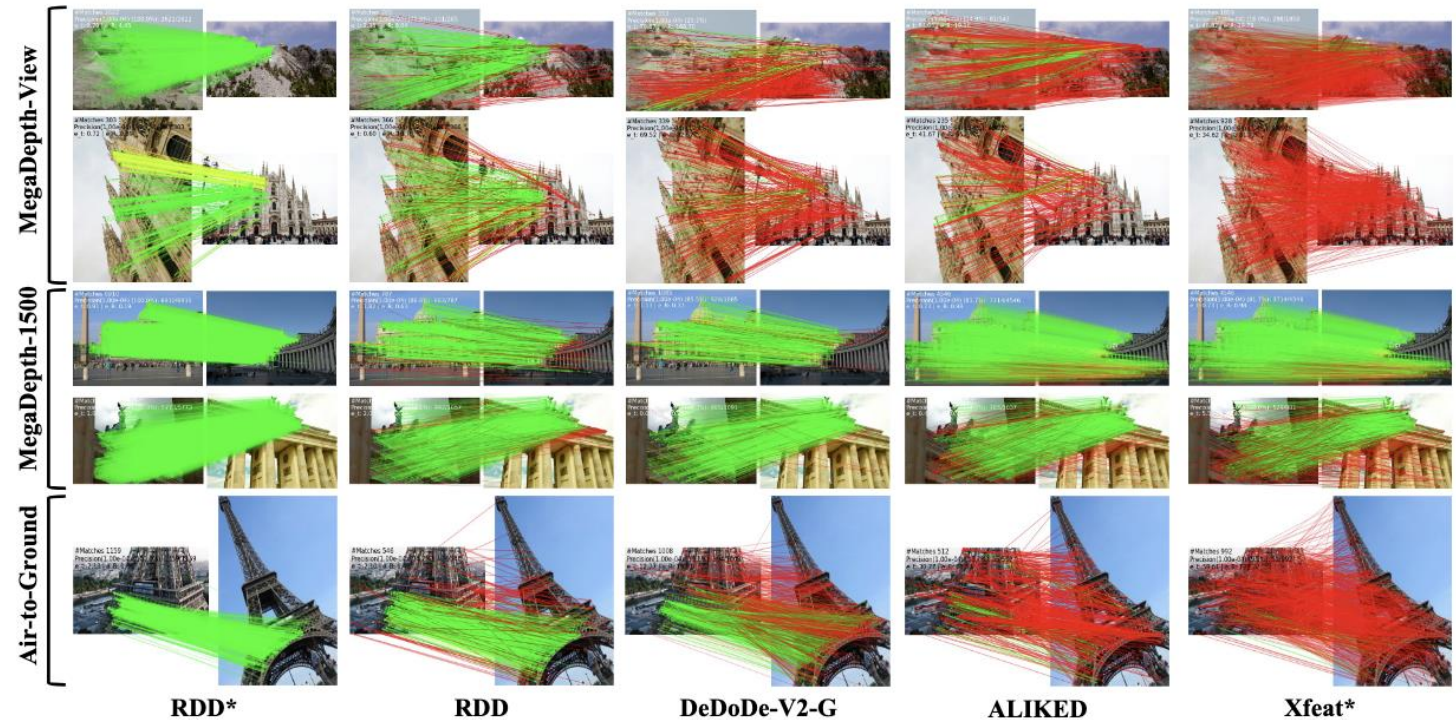


Figure 3. **Qualitative Results on MegaDepth.** RDD* and RDD are qualitatively compared to DeDoDe-V2-G [10, 11] ALIKED [47] and XFeat* [32]. RDD* and RDD are more robust compared DeDoDe-G* and ALIKED under challenging scenarios like large scale and viewpoint changes. The red color indicates epipolar error beyond 1×10^{-4} (in the normalized image coordinates)

빨간색: 정규화 이미지 좌표 기준 Epipolar Error가 1×10^{-4} 를 넘는 Match

DeDoDe-V2-G: DeDoDe v2 (Analyzing and Improving the DeDoDe Keypoint Detector), *: Semi-dense 버전 (RDD*, XFeat*)

● 4. Experiments – 4.2. Homography Estimation

Viewpoint 변화에서 RDD가 MHA@5-10px 1위, Illumination 변화에서는 기존 방법과 비슷한 성능

DATASET

- **HPatches**: Illumination 변화 Sequence 52개, Viewpoint 변화 Sequence 56개
- 짧은 변 480px로 Resize
- 추정 한 대응으로 **RANSAC** Homography 추정

METRICS & COMPARING

- XFeat-ALIKE를 따라 **MHA** 측정
- Reference 이미지 네 모서리를 GT와 추정 Homography로 각각 Target에 Warp → 평균 **Corner Error(px)**로 Accuracy 계산
- 비교: SuperPoint, DISK, ALIKED, DeDoDe-G, XFeat

RESULTS ON HPATCHES

- **Illumination**: 기존의 정확한 Detector/Descriptor와 비슷한 성능 (모든 방법이 잘 동작)
- **Viewpoint**: 다른 방법 대부분보다 우수
- Large Baseline Pair 같은 어려운 경우를 다루는 **Robustness** 근거

Table 3. **Homography estimation on HPatches**. All methods perform well for illumination sequences. RDD provides high quality homography estimation especially when there are significant viewpoint changes. Best in bold, second best underlined.

Method	Illumination			Viewpoint		
	@3px	MHA @5px	@10px	@3px	MHA @5px	@10px
SuperPoint [6] CVPRW'18	93.0	98.0	99.9	70.0	82.0	87.0
DISK [41] NeurIPS'20	96.0	98.0	98.0	72.0	81.0	84.0
ALIKED [47] TIM'23	97.0	99.0	99.0	78.0	<u>85.0</u>	<u>88.0</u>
DeDoDe-G [10] 3DV'24	<u>96.0</u>	99.0	99.9	68.0	77.0	80.0
XFeat [32] CVPR'24	90.0	96.0	98.0	55.0	72.0	80.0
RDD	93.0	99.0	99.9	<u>76.0</u>	86.0	90.0

Table 3 · Viewpoint, MHA@10px

RDD	90.0%
ALIKED	88.0%
SuperPoint	87.0%
DISK	84.0%
DeDoDe-G	80.0%
XFeat	80.0%

수치 읽는 법: Threshold 3·5·10px에서의 Homography Accuracy(%), 높을수록 좋음 · Bold = 1위, Underline = 2위 · Threshold는 Table 3 기준 (본문은 {3, 5, 7}px로 표기)

MHA: Mean Homography Accuracy, **GT**: Ground Truth, **RANSAC**: Random Sample Consensus

● 4. Experiments – 4.3. Visual Localization, 4.4. Ablation

Aachen Night에서 RDD가 세 Threshold 모두 1위, Ablation에서 모든 설계 요소 제거 시 성능 하락

4.3 VISUAL LOCALIZATION · DATASET

- Query 이미지의 Pose를 대응 **3D Scene Reconstruction** 기준으로 추정
- Aachen Day-Night**: 고품질 야간 이미지를 주간 3D Model에 Localize
- 14,607장, 날씨·계절·시점·Day-Night 변화

METRICS & COMPARING

- XFeat를 따라 **HLoc**으로 모든 방법 평가
- 긴 변 1,024px, Top 4,096 Keypoint
- Aachen Day-Night Benchmark의 평가 도구로 Translation {0.25m, 0.5m, 5m}, Rotation {2°, 5°, 10°} Threshold에서 Pose **AUC** 계산
- 비교: SuperPoint, DISK, ALIKE, ALIKED, XFeat (MNN Matching)

RESULTS

- 어려운 **Night** 설정에서 모든 방법보다 우수: **86.7 / 92.9 / 99.0**
- 결과가 RDD의 효과를 추가로 입증
- Day: 87.0 / 94.2 / 97.8

Table 4. **Visual Localization on Aachen day-night.** RDD outperforms previous state-of-the-art methods with MNN matching for more challenging night setting. Best in bold, second best underlined.

Methods	Day	Night	
	(0.25m, 2°) / (0.5m, 5°) / (1.0m, 10°)		
SuperPoint [6] CVPRW'18	87.4 / 93.2 / 97.0	<u>77.6</u>	85.7 / 95.9
DISK [41] NeurIPS'20	86.9 / 95.1 / 97.8	83.7	89.8 / 99.0
ALIKE [46] TM'22	85.7 / 92.4 / 96.7	81.6	88.8 / 99.0
ALIKED [47] TIM'23	86.5 / 93.4 / 96.8	<u>85.7</u>	91.8 / 96.9
XFeat [32] CVPR'24	84.7 / 91.5 / 96.5	77.6	89.8 / <u>98.0</u>
RDD	<u>87.0</u> / <u>94.2</u> / 97.8	86.7	92.9 / 99.0

4.4 ABLATION (Table 5, AUC@5°)

- 설계 요소를 바꾼 모든 Variant에서 Full 대비 하락
- Joint Training**: 한 Task만 실패해도 두 Branch가 갱신돼 Descriptor 수렴이 더 어려움
- Match Refinement** 제거: RDD* 51.3 → 41.3

Table 5. **Ablation study on MegaDepth.** We ablate the design choices for architecture and training strategies for relative pose estimation on MegaDepth.

Variant	AUC@5°	
	RDD	RDD*
Full	48.2	51.3
Larger patch size $s = 8$	44.6	49.4
Less sample points $N_{pq} = 4$	46.5	49.9
No keypoint branch	44.1	-
Joint training of two branches	42.8	46.9
No match refinement	-	41.3
Without Air-to-Ground Data	47.4	50.4

수치 읽는 법: Table 4는 (Translation, Rotation) Threshold별 Camera Pose AUC(%), Table 5는 MegaDepth Relative Pose AUC@5°, 높을수록 좋음 · Table 4 열 이름은 (1.0m, 10°), 본문은 5m로 표기
HLoc: Hierarchical Localization, **ALIKE**: Accurate and Lightweight Keypoint Detection and Descriptor Extraction, **s**: Patch Size, **N_{pq}**: Query당 Sampling Point 수, **A2G**: Air-to-Ground

● 5. Limitations and Conclusion

RDD는 기하 불변 특징을 효율적으로 추출했으나, **Sparse Prior** 의존과 **Data Augmentation** 부재가 한계

의의

- RDD는 **Deformable Transformer** 기반의 Robust한 Feature Detection·Description Framework로, **Sparse Image Matching**에서 뛰어나고 기하 변환과 다양한 조건에 강한 회복력을 보임
- **Deformable Attention**을 활용해 기하 불변 특징을 효율적으로 추출, 기존 방법 대비 **정확도와 적응성 향상**
- Keypoint Detection·Description의 의미 있는 진전으로, **Cross-view Scene Modeling**과 **Visual Localization** 같은 어려운 3D Vision Task 응용의 길을 엮

한계

- 학습 시 명시적 **Data Augmentation**이 없음
- **Sparse Correspondence**의 **Confident Prior**에 의존해 Semi-dense Matching이 실패할 수 있음
- 향후: Data Augmentation으로 학습하고 **Visual Feature**로 **Refinement Module**을 개선하면 성능 향상 가능

저자들은 이 연구가 미국 IARPA(Intelligence Advanced Research Projects Activity)의 지원을 받았다고 밝힘

Thank you

RDD: Robust Feature Detector and Descriptor using Deformable Transformer

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